

Week 1 • Day 6

Missing Value Handling & Data Cleaning

Fixed all data quality issues, standardised columns, and saved the clean dataset to PostgreSQL

DATE	INTERN NAME	PROJECT	MENTOR
Day 6 of 28 Week 1		Customer Churn Prediction & LTV Engine	

■ Technologies Used Today

Python

Pandas

NumPy

SQLAlchemy

PostgreSQL

■ Today's Key Metrics

7,043

Clean Rows Saved

Empty Strings Fixed

0

Missing Values Remaining

5

Binary Cols Encoded

■ Day Objectives

- Audit all data quality issues: NaN, empty strings, duplicates
- Fix TotalCharges 11 empty strings with MonthlyCharges imputation
- Remove duplicate rows by customerID
- Standardise column names to lowercase with underscores
- Convert Yes/No binary columns to 1/0 integers
- Save cleaned dataset to data/processed/ and PostgreSQL

■ Tasks Completed Today

1. Created `src/data_cleaning.py` with `audit_data_quality()` and `clean_data()` functions
2. Ran data audit — confirmed: 0 true NaN values, 11 empty strings in `TotalCharges`, 0 duplicates
3. Fixed `TotalCharges`: `pd.to_numeric(errors='coerce')` converted empty strings to NaN, then filled with `MonthlyCharges`
4. Verified the 11 rows with empty `TotalCharges` all have `tenure=0` — logically correct to use `MonthlyCharges`
5. Removed 0 duplicate rows (dataset was already clean in this regard)
6. Standardised all column names: `'customerID' → 'customerid'`, `'MonthlyCharges' → 'monthlycharges'`, etc.
7. Converted binary columns: `Partner`, `Dependents`, `PhoneService`, `PaperlessBilling`, `Churn` → 1/0
8. Ran assert statements to validate: no NaN in `TotalCharges`, no duplicates
9. Saved cleaned CSV to `data/processed/telco_churn_clean.csv`
10. Saved cleaned DataFrame to PostgreSQL as `customers_clean` table
11. Verified: `customers_clean` has 7,043 rows, 0 missing values
12. Committed all files to GitHub

■ Concepts Learned

- MCAR / MAR / MNAR — three categories of missing data with different handling strategies
- `pd.to_numeric(errors='coerce')` — converts invalid values to NaN gracefully
- Why imputing `TotalCharges` with `MonthlyCharges` is logically correct for `tenure=0` customers
- Assertion-based validation: using `assert` to enforce data quality guarantees
- Why standardising column names matters: avoids case-sensitivity bugs in SQL and Python
- Binary encoding: Yes/No → 1/0 is required for all sklearn models
- The difference between `if_exists='replace'` and `'append'` in `pandas to_sql()`

■ Tools / Software Used

Tool / Library	Version / Notes	Status
Pandas	2.1.4	✓ In use
NumPy	1.26.2	✓ In use
SQLAlchemy	2.0.23	✓ In use

■ Outputs / Deliverables Produced

- ◆ `src/data_cleaning.py` — full cleaning pipeline with audit + clean functions
- ◆ `data/processed/telco_churn_clean.csv` — 7,043 rows, 0 missing values
- ◆ `customers_clean` table in PostgreSQL — verified with pgAdmin
- ◆ 11 `TotalCharges` rows fixed (imputed with `MonthlyCharges`)
- ◆ All column names standardised to lowercase
- ◆ 5 binary columns converted to 1/0 integers
- ◆ GitHub commit with cleaning script and processed CSV

■ Problems Encountered & Solutions

Problem Faced	How I Solved It
assert statement failed: TotalCharges still had NaN	Checked code order — the fillna() was applied before to_numeric(). Swapped the order: to_numeric first, then fillna
Column names still uppercase after standardisation	Forgot to reassign: df.columns = [col.lower()...] must be saved back with df = df.copy() first

■ GitHub Commit Record

Commit Message	Week1-Day6: Data cleaning — fixed TotalCharges, standardised columns, saved clean dataset
Branch	main
Files Changed	src/data_cleaning.py, data/processed/telco_churn_clean.csv
Commit Hash	(fill after push)

■ End-of-Day Submission Checklist

■ data_cleaning.py created	■ Audit report printed
■ 11 TotalCharges rows fixed	■ 0 duplicates confirmed
■ Column names lowercased	■ 5 binary cols = 1/0
■ Assert statements pass	■ telco_churn_clean.csv saved
■ customers_clean in PostgreSQL	■ Day 6 commit on GitHub

■ Self Assessment

9/10 Self Rating	6 Hours Spent	Week 1, Day 6 Progress	6/28 Day in Program
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■ Key Learnings & Reflections

- ★ Always audit data BEFORE cleaning — understand what you're fixing and why
- ★ Assert statements are the best way to guarantee data quality at each step
- ★ Empty strings and NaN are different — must check for both in every dataset
- ★ Order of operations matters: convert to numeric THEN fill NaN, not the reverse

■ Plan for Tomorrow

- Generate a comprehensive EDA summary dashboard (multi-panel figure)
- Write `baseline_report.md` summarising all Week 1 insights
- Review all 7 daily commits on GitHub
- Prepare for Week 2: feature engineering plan

INTERN SIGNATURE	MENTOR SIGNATURE	DATE SUBMITTED